HZB BESSY II Light Source

BESSY II Datathon

Datathon Challenge: Tasks & Rules

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Sonal & Lukas

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Datathon Challenge

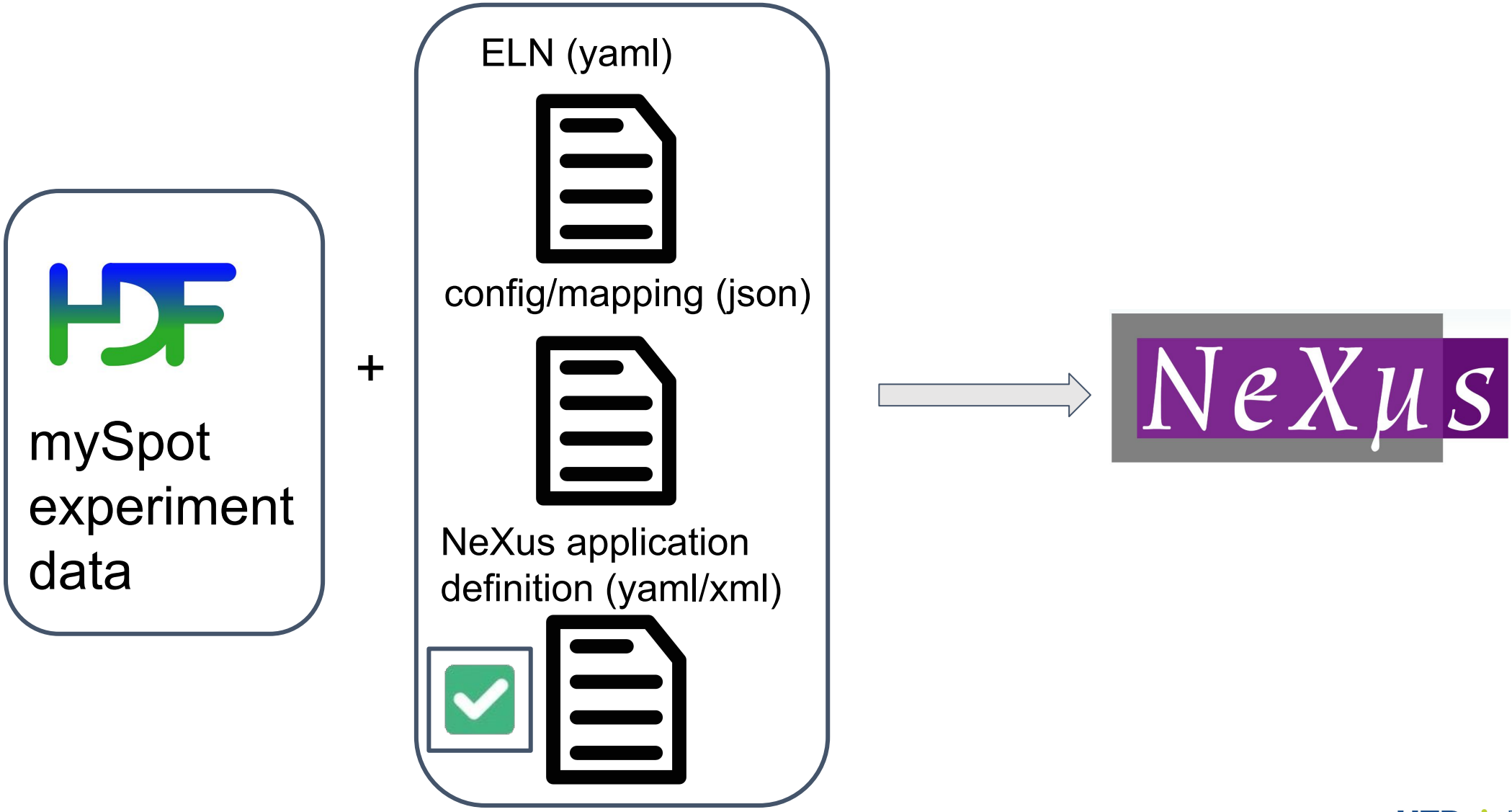
Datathon Challenge: Task & Rules

Overall Goal: Motivate and equip the participants with all tools needed to write and use reusable/reproducible data processing workflows

Challenge:

- Converting mySpot cuvette-scan experimental data into the NeXus format using pynxtools
  - Challenge 1: Populate NXdata & NXuser
  - Challenge 2: Convert into NXfluo
  - Challenge 3: Convert into NXmonopd
  - Challenge 4: Multimodal NeXus application definition
- Working on your data
  - Challenge 1: Check if NeXus application definition exist for your experiment data, populate at least three NeXus fields
  - Challenge 2: Write a custom NeXus application definition for your case (If NeXus application definition does not exist for your case)

Datathon Challenge



Datathon Challenge

Those having pynxtools running locally from Day 1:

- pynxtools >=v0.13.0
- config file
-
```
dataconverter
    --nxdl
    NXapp-def
    data.h5 eln.yaml
    -c config.json
    --output my_nexus.nxs
```

- work locally on your computer
- submit config file and output NeXus file (via git branch day2_YourName)

Those who do NOT have pynxtools working locally

- pynxtools <v0.13.0
- mapping file
-
```
dataconverter
    --reader json_map
    --nxdl NXapp-def
    --input_file data.h5
    --input_file eln.yaml
    --input_file xxx.mapping.json
    --output my_nexus.nxs
```

- work on jupyter notebook in NORTH (nubes folder)
- submit mapping file and jupyter notebook (via NOMAD upload share)